

# Quantum Tomography for High Energy Physics

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## 1 Quantum Tomography at work

### 1.1 One Spin-1/2 system

Let's describe the one spin-1/2 density matrix:

$$\rho = \frac{1}{2} (\mathcal{I} + \vec{B} \cdot \vec{\sigma}) \quad (1)$$

If we want to analyze a decay  $\tau/\bar{\tau} \rightarrow l^+/l^-$ , then the option of measure just two directions (up/down) isn't available, instead we need to measure in a space, in a solid angle space. So, we have the intention of measure in a specific direction in which the product of the decay goes on, say  $\hat{n}$ , then the projector operator will take the form:

$$\Pi_{\hat{n}} = \frac{1}{2} (\mathcal{I} + \hat{n} \cdot \vec{\sigma}) \quad (2)$$

The probability that the decay product be emitted in the  $\hat{n}$  direction is:

$$\begin{aligned} p(\hat{n}) &= \text{Tr} \left[ \frac{1}{2} (\mathcal{I} + \vec{B} \cdot \vec{\sigma}) \frac{1}{2} (\mathcal{I} + \hat{n} \cdot \vec{\sigma}) \right] \\ &= \text{Tr} \left[ \frac{1}{4} \left( \mathcal{I} + \vec{B} \cdot \vec{\sigma} + \hat{n} \cdot \vec{\sigma} + \underbrace{(\vec{B} \cdot \vec{\sigma})(\hat{n} \cdot \vec{\sigma})}_{(\vec{B} \cdot \hat{n})I + i(\vec{B} \times \hat{n}) \cdot \vec{\sigma}} \right) \right] \\ &= \frac{1}{4} \text{Tr} \left[ \mathcal{I} + \vec{B} \cdot \vec{\sigma} + \hat{n} \cdot \vec{\sigma} + (\vec{B} \cdot \hat{n})I + i(\vec{B} \times \hat{n}) \cdot \vec{\sigma} \right] \\ &= \frac{1}{4} \text{Tr} \left[ \mathcal{I} + (\vec{B} + \hat{n} + i(\vec{B} \times \hat{n})) \cdot \vec{\sigma} + (\vec{B} \cdot \hat{n})\mathcal{I} \right] \\ &= \frac{1}{4} \left[ \underbrace{\text{Tr} [\mathcal{I}]}_4 + \underbrace{\text{Tr} \left[ (\vec{B} + \hat{n} + i(\vec{B} \times \hat{n})) \cdot \vec{\sigma} \right]}_{\vec{\sigma}'\text{'s are traceless}} + \text{Tr} [(\vec{B} \cdot \hat{n})I] \right] \\ &= \frac{1}{4} \left[ 2 + 2(\vec{B} \cdot \hat{n}) \right] \end{aligned} \quad (3)$$

So, the probability is given by:

$$p(\hat{n}) = \frac{1}{2} \left[ 1 + \vec{B} \cdot \hat{n} \right] \quad (4)$$

However, the probability isn't normalized in the solid angle space. Let's say that  $P(\hat{n}) \propto 1 + \vec{B} \cdot \hat{n}$ , then:

$$\int d\Omega \, p(\hat{n}) = 1 \quad (5)$$

$$\int d\Omega \left[ c(1 + \vec{B} \cdot \hat{n}) \right] = 1 \quad (6)$$

$$c \underbrace{\int_{4\pi} d\Omega}_{4\pi} + c \vec{B} \cdot \underbrace{\int_0 d\Omega \, \hat{n}}_0 = 1 \quad (7)$$

$$c = \frac{1}{4\pi} \quad (8)$$

Finally, the probability density is given by:

$$p(\hat{n}) = \frac{1}{4\pi} \left[ 1 + \vec{B} \cdot \hat{n} \right] \quad (9)$$